3 **(a)** **(i)** The gravitational potential on the surface of a planet P of radius R is $-\phi_s$. The planet can be considered to be an isolated sphere.

On the Fig. 3.1, sketch the variation of the gravitational potential with distance r from the centre of the planet for values of r between R and $4R$.

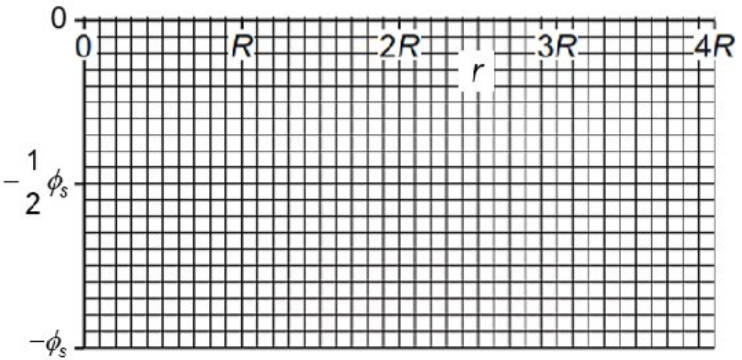


Fig. 3.1

[1]

(ii) Explain why is the gravitational potential negative.

.....

.....

..... [2]

(b) The variation with $\frac{1}{r}$ of the gravitational potential ϕ at different distance r from the centre of planet P is shown in Fig. 3.2.

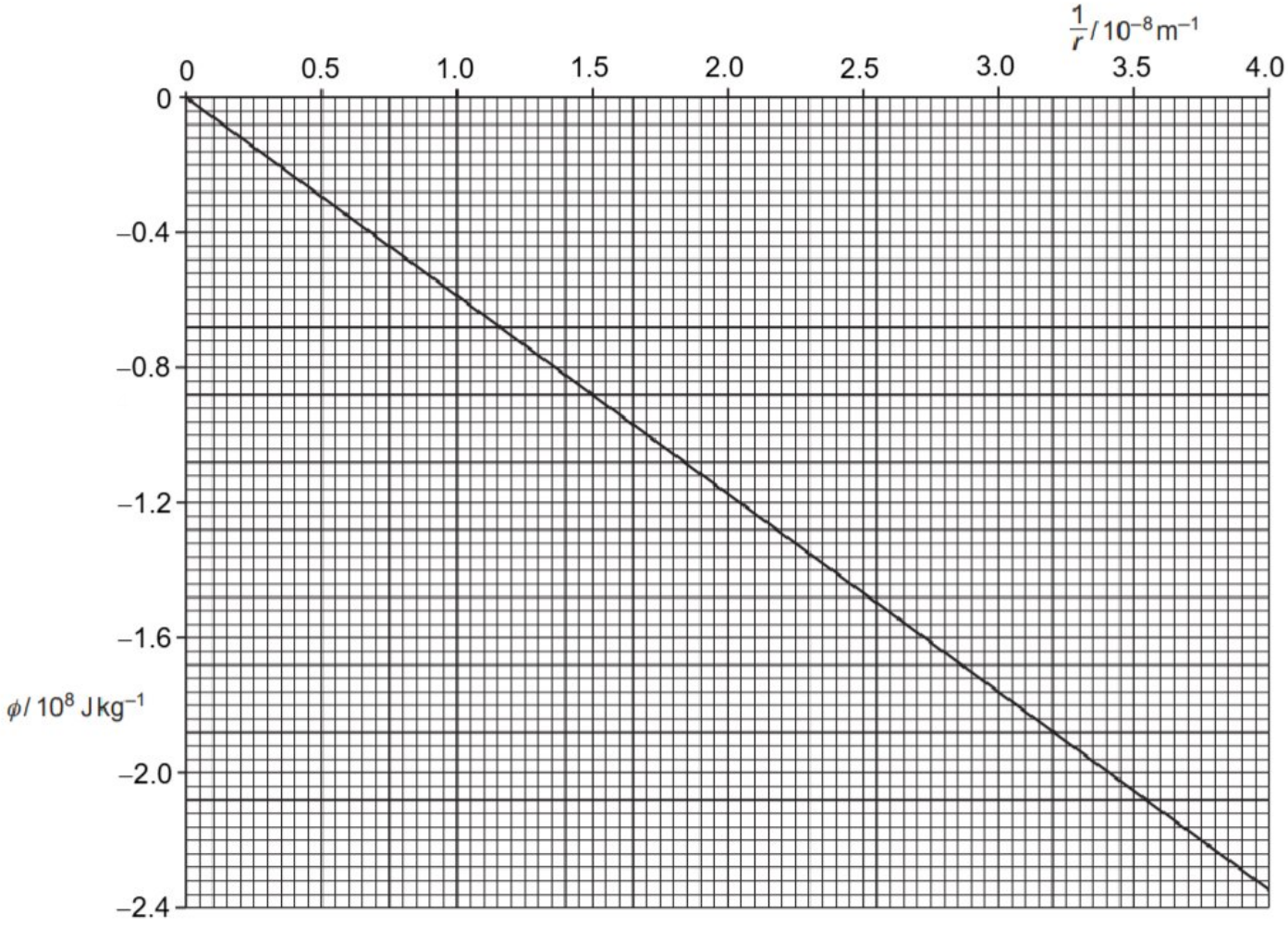


Fig. 3.2

Using Fig. 3.2, show that the mass of the planet is 8.8×10^{25} kg.

[1]

(c) **(i)** The period of rotation of the planet is 0.72 Earth days.

A satellite in orbit around the planet remains above the same point on the surface of the planet.

Determine the radius of the orbit of the satellite. Explain your working clearly.

radius = ... m [3]

(ii) The speed of the satellite in **(c)(i)** is 8400 m s^{-1}. The mass of the satellite is 1200 kg. Determine the minimum amount of additional energy required to move the satellite from its orbit to infinity.

energy = ... J [3]

[Total: 10]